

Report

Aim: To segment five brain tissue regions from the provided 10 T1-weighted MRI images, using 2D and 3D segmentation algorithms, evaluating their performance and to compare their performance against the ground-truth labels to discover which algorithm is the most successful.

General Preprocessing(used for all tasks) – Figure 1

For all experiments I first applied the same pre-processing pipeline to the entire volume so that every algorithm sees identical input data.

Global Intensity normalization: The original T1 volume (T1, size 362×434×10) was converted to float32 and normalised globally to the range [0,1] using the overall minimum and maximum intensity across all slices. This contrast stretching makes the intensity range comparable between slices and stabilises thresholding and clustering.

Foreground Masking(K = 2) and Morphological Cleaning: Using K-means with K=2 on all voxels of the normalised volume (intensity only). The cluster with the lower mean intensity corresponds to air/background; the other cluster is taken as foreground. This initial mask was then cleaned per slice using simple **scipy.ndimage** operations: binary opening and closing with a 5×5 structuring element, hole filling, and connected-component analysis where only the largest component was kept as the brain. The result is a smooth binary mask that isolates brain tissue and removes skull/air. All later algorithms operate only inside this mask and any voxel outside it is forced to label 0 (background). *Figure 1* shows a representative slice of the normalised T1 image and its cleaned foreground mask.

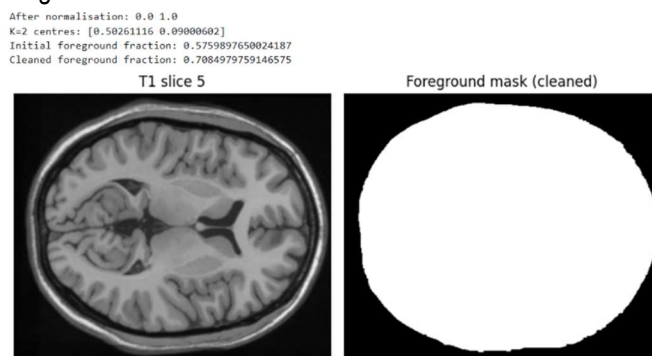


Figure 1.

Cluster to Label Remapping: To compare unsupervised segmentations with the ground truth, all methods share the same cluster-to-label remapping step. For each slice and each cluster index, I look up the ground-truth labels of all voxels assigned to that cluster and assign the cluster the label (1–5) that occurs most frequently. Voxels outside the mask are always mapped to label 0. This ensures that each cluster is interpreted as one of the tissue labels when computing evaluation metrics.

TASK 1 – 2D Tissue Segmentation

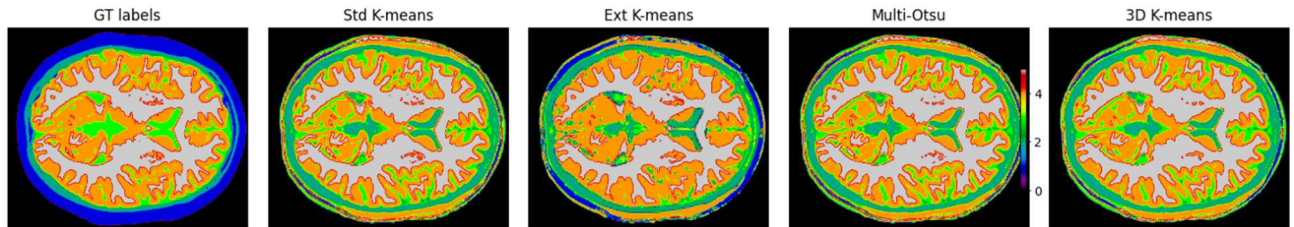
Aim: To discover which 2D segmentation algorithm is best for determining tissue regions across all T1 weighted MRI slices

Method 1: Standard K-means(Intensity only) – For each 2D slice of the normalised volume I applied **K-means clustering** to foreground voxels only, using **sklearn.cluster.KMeans** with K = 5 clusters(one per tissue class), n_init=5, max_iter=150 and a fixed random seed for reproducibility. The feature vector for each voxel(now pixel) is just its normalised intensity. After clustering, the per-slice cluster-to-label majority mapping it converts cluster indices into labels 1–5, while background pixel remains 0. The algorithm and parameters are identical for all slices, only the input intensities change.

Method 2: Multi-Otsu Thresholding – The second 2D method is **multi-level Otsu thresholding** on each slice, applied only within the brain mask. For each slice I use `skimage.filters.threshold_multiotsu` with `classes = 5` on the foreground intensities to obtain four thresholds that partition the histogram into five intensity ranges. Using `numpy.digitize`, then every voxel in the slice is assigned to one of five bins (0–4) based purely on intensity. These bins are then treated as five clusters and mapped to labels 1–5 by the same majority-vote scheme. This gives a purely histogram-based, parameter-free segmentation that is again applied identically to each slice.

Method 3: Extended K-means (Intensity + Scaled Sobel Gradients) – The third 2D method extends K-means with an additional edge feature. For each slice I compute horizontal and vertical Sobel filters using `skimage.filters.sobel`, combine them into a gradient magnitude image, and normalise this gradient to 0,1. For each foreground voxel I form a 2D feature vector : $[I_{norm}, \alpha \cdot |\nabla I|]$, where α scales the influence of edges.

I then run K-means on these 2D features with K in $\{5, 6, 7, 8\}$ and α in $\{0.1, 0.2, 0.3, 0.5, 0.7\}$. For every (K, α) pair, clustering is repeated on every slice with the same parameters, followed by the same cluster-to-label mapping. This satisfies the “same algorithm per slice” requirement while allowing me to tune K and α globally.



The above figure is a multi-panel image showing, for the same slice(5): GT labels, Standard K-means, Extended K-means, Multi-Otsu and 3D Volume K-means(to be explored in Task 3).

Task 2 – Result Evaluation

Aim: To evaluate which segmentation algorithm performs best using reliable evaluation metrics.

Metrics Used: For the comparison of the segmentation Dice coefficients and Accuracy were used. Dice directly measures overlap between segmented and ground-truth tissues and is widely used in medical image segmentation, making it a suitable primary metric here. Accuracy complements Dice by reflecting pixel-level correctness and showing if a method is over- or under-segmenting background.

These metrics are implemented once in `compute_metrics` and applied consistently to every algorithm and shown below.

```

==== Standard K-means 2D (intensity only, K=5) ====
Per-class Dice (0..5): [0.9649 0.2922 0.7518 0.3383 0.813  0.9195]
Mean Dice 0-5: 0.6799
Mean Dice 1-5: 0.6229
Accuracy 0-5: 78.93 %
Accuracy 1-5 (GT>0): 71.06 %

==== Multi-Otsu 2D (classes=5, FG only) ====
Per-class Dice (0..5): [0.9649 0.275  0.7509 0.386  0.8271 0.9428]
Mean Dice 0-5: 0.6911
Mean Dice 1-5: 0.6364
Accuracy 0-5: 79.83 %
Accuracy 1-5 (GT>0): 72.3 %

==== Extended K-means 2D (intensity + scaled Sobel) ====
Per-class Dice (0..5): [0.9649 0.2922 0.7518 0.3383 0.813  0.9195]
Mean Dice 0-5: 0.6799
Mean Dice 1-5: 0.6229
Accuracy 0-5: 78.93 %
Accuracy 1-5 (GT>0): 71.06 %

==== 3D K-means 2D (intensity + scaled Sobel) ====
Per-class Dice (0..5): [0.9649 0.2922 0.7518 0.3383 0.813  0.9195]
Mean Dice 0-5: 0.6799
Mean Dice 1-5: 0.6229
Accuracy 0-5: 78.93 %
Accuracy 1-5 (GT>0): 71.06 %

```

Multi-Otsu slightly outperforms standard K-means in both Dice and Accuracy, indicating that global histogram thresholds provide a strong baseline when only intensity is used. However, the extended 2D K-means clearly gives the best overall performance: it achieves the highest mean Dice over tissue labels (0.713) and the highest

foreground accuracy (75.53%). This improvement is due to the additional gradient feature, which helps distinguish boundaries between tissues with similar intensities and reduces misclassification around edges.

Task 3 – Advanced 3D Tissue Segmentation

Aim: To explore whether using 3D segmentation with information across all slices improves segmentation accuracy compared to 2D slice methods.

Method: 3D K-means on the Volume (Intensity Only + 3D Majority Filter) – For the 3D method I treat the entire brain volume as a single dataset rather than independent slices. All foreground voxels from all slices are stacked into one list and clustered using K-means on intensity only.

To choose the number of clusters, I ran K-means for K in {5, 6, 7, 8} and recorded the inertia. Using a simple elbow heuristic on these inertias, the largest drop occurred when moving from K=4 to K=5, so I selected K = 5 as the optimal number of clusters for the full volume. I then ran K-means once more with K=5, n_init=5 and max_iter=200 on all foreground voxels and reshaped the resulting labels back into the 3D shape.

Cluster indices were converted to tissue labels using the same majority mapping per slice as in Task 1, and background voxels remained at label 0. To incorporate 3D spatial consistency, I applied a 3×3×3 majority filter using **scipy.ndimage.generic_filter** over the label volume: each voxel's label is replaced by the most frequent label in its 3D neighbourhood. This removes small, isolated regions and enforces smoother tissue regions across slices while still being a simple, interpretable post-processing step.

```
3D K-means elbow Ks: [5, 6, 7, 8]
3D K-means inertias: [1833.86 1303.1 901.76 701.15]
Chosen K for 3D volume: 5
=== 3D K-means volume (intensity, K=5) ===
Per-class Dice (0..5): [0.9626 0.2892 0.7389 0.3938 0.8358 0.941 ]
Mean Dice 0-5: 0.6935
Mean Dice 1-5: 0.6397
Accuracy 0-5: 80.14 %
Accuracy 1-5 (GT>0): 72.73 %
```

===== SUMMARY (labels 0-5 and 1-5) =====

Std K-means 2D	Dice0-5: 0.680	Dice1-5: 0.623	Acc0-5: 78.93%	Acc1-5: 71.06%
Ext K-means 2D	Dice0-5: 0.755	Dice1-5: 0.713	Acc0-5: 82.18%	Acc1-5: 75.53%
Multi-Otsu 2D	Dice0-5: 0.691	Dice1-5: 0.636	Acc0-5: 79.83%	Acc1-5: 72.30%
3D K-means vol	Dice0-5: 0.694	Dice1-5: 0.640	Acc0-5: 80.14%	Acc1-5: 72.73%

The above figure shows the dice and accuracy for 3D K-means and a summary of all the techniques used and their metrics respectively.

Conclusion

Compared to the purely intensity-based 2D methods, the 3D algorithm gives slightly higher mean Dice and accuracy than standard K-means and Multi-Otsu. This indicates that clustering all slices together and then smoothing with a 3D majority filter improves consistency across slices, reducing slice-to-slice variability and isolated misclassifications. In the multi panel image far above the 3D result appears smoother and more coherent around deep structures compared to the standard 2D methods.

However, the extended 2D K-means remains the top performer overall: its mean Dice over tissue labels (0.713) and foreground accuracy (75.53%) are higher than those of the 3D method. The main reason is that the 3D algorithm uses intensity only, whereas extended K-means benefits from both intensity and gradient features, which better separate tissues with overlapping intensities. In addition, forcing a single set of global cluster centres on the whole volume can be sub-optimal for slices whose intensity distributions deviate from the average, leading to small local mismatches that 3D smoothing cannot fully fix.

In summary, the 3D K-means method demonstrates that incorporating 3D information and simple majority filtering can yield more spatially consistent segmentations and slightly better performance than the basic 2D intensity-based methods. Nevertheless, the best overall segmentation algorithm is the extended 2D K-means with intensity + scalable sobel gradient features, which achieves the highest Dice and accuracy while still using straightforward classical techniques.